

Caden Sun

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SKILLS

Languages: Python, JavaScript/TypeScript, HTML/CSS, C/C++, SQL

Tools/Frameworks: PyTorch, Pandas, NumPy, Matplotlib, React, Node.js, Azure, Linux, Docker

EDUCATION

University of Waterloo

Sep 2025 - Present

Bachelor of Mathematics

EXPERIENCE

AI Intern | *ProCogia*

May 2026 - Present

- Supporting the development and **end-to-end deployment** of AI solutions for client companies, including API integration, cloud infrastructure considerations, service orchestration, and production monitoring.
- Working with **AWS services** such as **S3**, **EC2**, and **IAM** to implement cloud infrastructure, permissions, storage, and compute workflows.
- Conducting research for end-to-end voice AI deployments targeting **sub-500ms perceived latency**, using a modular stack involving **LiveKit**, **Deepgram Nova-3**, **Groq**, **Cartesia Sonic-3**, and **Krisp**.

ML Engineer | *Wat Street (UWaterloo's Quantitative Finance Team)*

Jan 2025 - Present

- Working on a **quantitative research project** analyzing how price and volatility changes in one stock propagate across correlated equities in the S&P 500.
- Engineered a data pipeline that processes S&P 500 market data into aligned snapshots using **Pandas** and **NumPy**, allowing for (30×500) matrices to be fed into input tensors.
- Developed and benchmarked baseline volatility forecasters (**HAR-RV**, **GARCH(1,1)**, **LSTM**) with evaluation using **RMSE** and **QLIKE**.
- Collaborating with teammates to validate results and **debug models** based on empirical performance.

Software Developer | *Waterloo Aerial Robotics Group (WARG)*

Oct 2025 - Apr 2026

- Designing autonomous aerial systems using **Python**, focusing on **system control** and **software design**.
- Building internal tools and mission interfaces using **Dart** and **Flutter** to enhance system monitoring.
- Added backend functionality (**JavaScript**) such as a coordinate mapping system to **IMACS-3.0**, WARG's custom ground control station for drones.
- Gained hands-on experience in **robotics software** workflows, cross-platform development, and **Git**.

PROJECTS

AI Chess Bot | *Python, PyTorch, Machine Learning* | 🌐

- Developed a self-learning chess engine in **PyTorch** that evaluates board states using a three layer **neural network** ($768 \rightarrow 256 \rightarrow 128 \rightarrow 1$) and **ReLU activations**.
- Created a training dataset** by generating thousands of randomized legal chess board positions through self-play and random move sampling.
- Calculated **training loss** by comparing the neural network's board state evaluations directly against Stockfish evaluation scores.
- Optimized the model using **Adam** and **MSE loss** to minimize deviation from Stockfish's evaluations, improving stability across varied game states.

Hearth | *Python, React, TypeScript* | 🌐

- Contributed on a **web platform** that **helps elderly and disabled homebuyers** estimate accessibility-focused renovation costs before purchasing a home.
- Implemented functionality allowing users to input a house listing URL which extracts property details for analysis.
- Integrated the **Gemini Flash API** to generate AI-assisted before-and-after renovation visualizations, illustrating potential accessibility upgrades such as ramps, widened doorways, and bathroom modifications.
- Developed a **Python** backend to handle data processing and AI requests, with a **React** and **TypeScript** frontend for a user-friendly interface.

Stock Screener | *Python, Pandas, NumPy, React, TypeScript* | 🌐

- Built a stock analysis and screening tool that visualizes market data in real time using the **Yahoo Finance API**.
- Implemented data-driven filtering algorithms by leveraging **Pandas** and **NumPy Python libraries** to identify stocks meeting specific criteria such as trading volume, call/put options, and volatility patterns.
- Developed a **React-based** frontend with **Recharts** to present a clean and visually pleasing **UI** along with interactive graphs.